Regime detection that can say what happened — and a measurement of how much that is worth

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> **The claim.** A two-state HMM refitted inside every walk-forward fold separates SPY volatility regimes in 18 of the 20 folds containing both states (sign test **p = 0.00040**). Reading only news published *before* each of 20 transitions, a language model produced cited explanations that match back to their own fortnight **72% of the time against 33% chance**, with **96.4% of 474 claims grounded and zero fabricated citations**. Whether they are *diagnostic* of a regime change is unresolved: **+18.5pp over matched controls, 95% CI [-12, 47]pp**.

PROBLEM

A regime model tells you the market changed but not what happened, because it only ever sees returns. Its output is coloured bands a human interprets from memory — undocumented, unauditable, different for everyone who looks. For a platform running thirty portfolio managers, that is thirty private readings of one signal.

SOLUTION

The HMM decides *when* the regime changed; the language model only describes *what was in the news* then. No statistic comes from the language model. For each date the tool retrieves a 14-day window that *provably* closes on it, and returns an explanation whose every claim cites a supplied article.

The reusable part is the **controls**, not the explanations. Anyone can ask a model what happened in March 2020; nobody can otherwise tell you whether the answer is specific to that window or boilerplate fitting any month. Point the tool at any regime model's dates and each explanation arrives with a blind-match score, a grounding rate, and a matched control arm.

IMPACT & VALUE

Stressed / calm volatility	Same-day	Forward 20d
SPY *(fitted)*	**2.18×**	**1.69×**
Nikkei 225	1.54×	1.18×
Hang Seng	1.46×	1.32×
DAX	1.87×	1.58×

A ratio of 2.18× means the stressed state's daily moves are 2.18 times the size of the calm state's. **Same-day** is partly definitional — the HMM is fed trailing volatility, so it had better separate on it. **Forward-20d** measures volatility that had *not happened* when the state was assigned, and is the honest number. Nikkei at 1.18× is close to nothing, and is reported as such. Within SPY the per-fold median is 1.52×, below the pooled figure because pooling mixes calm and crisis years.

USE OF AI

`claude-opus-5` via the Anthropic Messages API — one call per window, effort high, schema enforced server-side. Prompts are version-controlled files with a dated iteration history, never inline strings; every call logs model id, prompt hash and input hash. Built with Claude Code. News via the revision-pinned Wikipedia MediaWiki API, prices via yfinance, statistics via hmmlearn and scikit-learn, tool in Streamlit.

THREE CONTROLS

- **Hindsight.** A window object refuses to construct if any item post-dates it: 60 windows, 10,888 items, zero survivors. Pages are pinned by revision id because a mean of **38.2%** of a current Wikipedia day-page (peak 71.2%) was written *after* the boundary.
- **Memorisation.** Dates stripped; every claim must cite; grounding scored lexically, so the check has no world knowledge.
- **Placebo.** Era-matched controls, identical prompt and pipeline.

REFLECTIONS

The placebo arm went against me and is the result I would keep. Calling the explanations "not diagnostic" would be a no-difference claim that p = 0.43 does not license. **Power at the observed gap is 0.17** — one chance in six of detecting the effect observed. The binding constraint is 20 transitions, not controls, so more placebos cannot fix it; pre-specifying volatility *onsets* would roughly triple it.

> **What replicates.** Re-run on `claude-sonnet-5`, blind matching is identical and grounding differs by 0.4pp — properties of having the news. The confidence label is not (κ = 0.25; Sonnet declined half as often), which is why blind matching is load-bearing and the self-report is a weak instrument.

Given more time: more transitions, not more controls. One leak stays open — the boundary is 23:59 UTC and Wikipedia reports closes the same evening — though controls carry *more* of it than transitions, so it does not manufacture the gap.